

EXPERIMENT -3

AIM

To create an Employee–Department–Project relational database, insert at least 30 employees across 5 departments and 8 projects, and demonstrate SQL queries using **Selection, Projection, Aggregate Functions, GROUP BY, HAVING, CASE expressions, and ORDER BY**.

SOURCE CODE

```
CREATE DATABASE CompanyDB;
USE CompanyDB;
CREATE TABLE Department (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50) NOT NULL
);
CREATE TABLE Project (
    project_id INT PRIMARY KEY,
    project_name VARCHAR(100) NOT NULL,
    budget DECIMAL(12,2),
    dept_id INT,
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);
CREATE TABLE Employee (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50) NOT NULL,
    salary DECIMAL(10,2),
    hire_date DATE,
    dept_id INT,
    project_id INT,
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id),
    FOREIGN KEY (project_id) REFERENCES Project(project_id)
);
```

Insert Departments

```
INSERT INTO Department VALUES
(1,'IT'),
(2,'HR'),
(3,'Finance'),
(4,'Marketing'),
(5,'Operations');
```

Insert Projects

```
INSERT INTO Project VALUES
(101,'Cloud Migration',150000,1),
(102,'AI Analytics',200000,1),
(103,'Recruitment Portal',80000,2),
(104,'Financial Dashboard',120000,3),
```

(105,'Digital Campaign',90000,4),
(106,'Supply Chain System',180000,5),
(107,'Mobile Application',140000,1),
(108,'Employee Wellness',60000,2);

Insert 30 Employees

```
INSERT INTO Employee VALUES  
(1,'Aarav',65000,'2022-01-10',1,101),  
(2,'Vivaan',72000,'2021-03-15',1,102),  
(3,'Aditya',58000,'2023-06-20',1,107),  
(4,'Arjun',81000,'2020-08-12',1,101),  
(5,'Kabir',69000,'2022-11-05',1,102),  
(6,'Reyansh',62000,'2024-02-18',1,107),  
  
(7,'Ananya',52000,'2022-04-11',2,103),  
(8,'Diya',48000,'2023-01-22',2,108),  
(9,'Myra',61000,'2021-07-19',2,103),  
(10,'Sara',55000,'2022-09-30',2,108),  
(11,'Ishita',47000,'2024-03-14',2,103),  
(12,'Meera',59000,'2020-12-01',2,108),  
  
(13,'Rohan',75000,'2021-02-17',3,104),  
(14,'Karan',68000,'2022-05-09',3,104),  
(15,'Nikhil',83000,'2019-10-21',3,104),  
(16,'Yash',62000,'2023-08-13',3,104),  
(17,'Manav',71000,'2021-11-28',3,104),  
(18,'Dev',56000,'2024-01-09',3,104),  
  
(19,'Aanya',54000,'2022-02-25',4,105),  
(20,'Kiara',63000,'2021-06-16',4,105),  
(21,'Tanya',57000,'2023-04-10',4,105),  
(22,'Riya',66000,'2020-09-07',4,105),  
(23,'Avni',51000,'2024-02-05',4,105),  
(24,'Navya',70000,'2022-12-19',4,105),  
  
(25,'Samar',60000,'2021-01-12',5,106),  
(26,'Dhruv',73000,'2020-04-23',5,106),  
(27,'Atharv',67000,'2022-07-18',5,106),  
(28,'Parth',59000,'2023-05-29',5,106),  
(29,'Rudra',76000,'2019-08-31',5,106),  
(30,'Veer',64000,'2024-01-20',5,106);
```

SQL QUERIES

1. Selection

Display employees whose salary is greater than 70,000.

```
SELECT *
```

```
FROM Employee
```

```
WHERE salary > 70000;
```

OUTPUT

 AKHIL VERMA

I/O

AI Agent

83 ms 

STDIN

Input for the program (Optional)

Output:

emp_id	emp_name	salary	hire_date	dept_id	project_id
2	Vivaan	72000.00	2021-03-15	1	102
4	Arjun	81000.00	2020-08-12	1	101
13	Rohan	75000.00	2021-02-17	3	104
15	Nikhil	83000.00	2019-10-21	3	104
17	Manav	71000.00	2021-11-28	3	104
26	Dhruv	73000.00	2020-04-23	5	106
29	Rudra	76000.00	2019-08-31	5	106

2. Projection

Display only employee names and salaries.

```
SELECT emp_name, salary  
FROM Employee;
```

OUTPUT



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AI Agent

95 ms



STDIN

Input for the program (Optional)

Output:

```
+-----+-----+
| emp_name | salary  |
+-----+-----+
| Aarav    | 65000.00 |
| Vivaan   | 72000.00 |
| Aditya   | 58000.00 |
| Arjun    | 81000.00 |
| Kabir    | 69000.00 |
| Reyansh  | 62000.00 |
| Ananya   | 52000.00 |
| Diya     | 48000.00 |
| Myra     | 61000.00 |
| Sara     | 55000.00 |
| Ishita   | 47000.00 |
| Meera    | 59000.00 |
| Rohan    | 75000.00 |
| Karan    | 68000.00 |
```

3. Aggregate Functions

Find total employees, average salary, maximum salary, minimum salary and total salary.

```
SELECT
  COUNT(*) AS total_employees,
  AVG(salary) AS average_salary,
  MAX(salary) AS highest_salary,
  MIN(salary) AS lowest_salary,
  SUM(salary) AS total_salary
FROM Employee;
```

OUTPUT:



AKHIL VERMA

I/O

AI Agent

86 ms

STDIN

Input for the program (Optional)

Output:

total_employees	average_salary	highest_salary	lowest_salary	total_salary
30	63466.666667	83000.00	47000.00	1904000.00

4. GROUP BY

Display the number of employees and average salary in each department.

```
SELECT
    d.dept_name,
    COUNT(e.emp_id) AS employee_count,
    AVG(e.salary) AS average_salary
FROM Department d
JOIN Employee e
ON d.dept_id = e.dept_id
GROUP BY d.dept_id, d.dept_name;
```

OUTPUT



AKHIL VERMA

I/O

AI Agent

105 ms

STDIN

Input for the program (Optional)

Output:

dept_name	employee_count	average_salary
IT	6	67833.333333
HR	6	53666.666667
Finance	6	69166.666667
Marketing	6	60166.666667
Operations	6	66500.000000

5. HAVING

Display departments whose average salary is greater than 65,000.

```
SELECT
    d.dept_name,
    AVG(e.salary) AS average_salary
FROM Department d
JOIN Employee e
ON d.dept_id = e.dept_id
GROUP BY d.dept_id, d.dept_name
HAVING AVG(e.salary) > 65000;
```

OUTPUT

Department	Average Salary
------------	----------------

IT	67833.33
----	----------

Finance	69166.67
---------	----------

Operations	66500.00
------------	----------

6. CASE Expression

Classify employees according to their salary.

```
SELECT
    emp_name,
    salary,
    CASE
        WHEN salary >= 75000 THEN 'High Salary'
        WHEN salary >= 60000 THEN 'Medium Salary'
        ELSE 'Low Salary'
    END AS salary_category
FROM Employee;
```

OUTPUT



AKHIL VERMA

I/O

AI Agent

94 ms

STDIN

Input for the program (Optional)

Output:

```
+-----+-----+-----+
| emp_name | salary  | salary_category |
+-----+-----+-----+
| Aarav    | 65000.00 | Medium Salary   |
| Vivaan   | 72000.00 | Medium Salary   |
| Aditya   | 58000.00 | Low Salary      |
| Arjun    | 81000.00 | High Salary     |
| Kabir    | 69000.00 | Medium Salary   |
| Reyansh  | 62000.00 | Medium Salary   |
| Ananya   | 52000.00 | Low Salary      |
| Diya     | 48000.00 | Low Salary      |
| Myra     | 61000.00 | Medium Salary   |
| Sara     | 55000.00 | Low Salary      |
| Ishita   | 47000.00 | Low Salary      |
| Meera    | 59000.00 | Low Salary      |
| Rohan    | 75000.00 | High Salary     |
| Karan    | 68000.00 | Medium Salary   |
| Nikhil   | 83000.00 | High Salary     |
```

...

7. ORDER BY

Display employees in descending order of salary.

```
SELECT emp_name, salary
```

```
FROM Employee
```

```
ORDER BY salary DESC;
```

OUTPUT



AKHIL VERMA

I/O

AI Agent

86 ms

STDIN

Input for the program (Optional)

Output:

```
+-----+-----+
| emp_name | salary  |
+-----+-----+
| Nikhil   | 83000.00 |
| Arjun    | 81000.00 |
| Rudra    | 76000.00 |
| Rohan    | 75000.00 |
| Dhruv    | 73000.00 |
| Vivaan   | 72000.00 |
| Manav    | 71000.00 |
| Navya    | 70000.00 |
| Kabir    | 69000.00 |
| Karan    | 68000.00 |
| Atharv   | 67000.00 |
| Riya     | 66000.00 |
| Aarav    | 65000.00 |
| Veer     | 64000.00 |
| Kiara    | 63000.00 |
```

8. JOIN Employee, Department and Project

Display employee name, department, project and salary.

```
SELECT
    e.emp_name,
    d.dept_name,
    p.project_name,
    e.salary
FROM Employee e
JOIN Department d
ON e.dept_id = d.dept_id
JOIN Project p
ON e.project_id = p.project_id
ORDER BY d.dept_name, e.emp_name;
```

OUTPUT





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I/O
AI Agent
83 ms 

STDIN

Input for the program (Optional)

Output:

emp_name	dept_name	project_name	salary
Dev	Finance	Financial Dashboard	56000.00
Karan	Finance	Financial Dashboard	68000.00
Manav	Finance	Financial Dashboard	71000.00
Nikhil	Finance	Financial Dashboard	83000.00
Rohan	Finance	Financial Dashboard	75000.00
Yash	Finance	Financial Dashboard	62000.00
Ananya	HR	Recruitment Portal	52000.00
Diya	HR	Employee Wellness	48000.00
Ishita	HR	Recruitment Portal	47000.00
Meera	HR	Employee Wellness	59000.00
Myra	HR	Recruitment Portal	61000.00
Sara	HR	Employee Wellness	55000.00
Aarav	IT	Cloud Migration	65000.00
Aditya	IT	Mobile Application	58000.00
Ariun	IT	Cloud Migration	81000.00

RESULT

The **Employee–Department–Project database** was successfully created with:

- **30 Employees**
- **5 Departments**
- **8 Projects**

The following SQL concepts were successfully demonstrated:

1. Selection
2. Projection
3. Aggregate Functions
4. GROUP BY
5. HAVING
6. CASE Expression
7. ORDER BY
8. JOIN